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M1 MACRO

Group A

Session 15

These notes are written during class. They are posted "for the record" and may contain mistakes. They are not intended to be read by anyone that did not attend the class.

Reciprocal Result

• If λ_1, λ_2 are the roots of $a\lambda^2 + b\lambda + c = 0$ (1)
($\neq 0$)

Then, the roots of $a + b\lambda + c\lambda^2 = 0$

are $\frac{1}{\lambda_1}$ and $\frac{1}{\lambda_2}$

Proof define μ s.t. $\lambda = \frac{1}{\mu}$, $\mu \neq 0$

$$(1) \quad \frac{a}{\mu^2} + \frac{b}{\mu} + c = 0 \quad \times \mu^2$$

$$a + b\mu + c\mu^2 = 0 \Rightarrow \text{roots are } \frac{1}{\lambda_1}, \frac{1}{\lambda_2}$$